MILITARY SPECIFICATION

CHART SET, VISION ACUITY TESTING, Clinical and Screening, Distant Vision

1. SCOPE

1.1 This specification covers a set of three alternate forms of charts for clinical and screening use in testing distant vision acuity.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585 Boxes, Wood, Wirebound.
PPP-B-601 Boxes, Wood, Cleated-Plywood.
PPP-B-621 Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 Box, Fiberboard.

Military

MIL-L-10547 Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6515

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

TAPPI STANDARDS

T 401 M-53 Fiber Analysis of Paper and Paperboard.
T 402 M-49 Conditioning Paper and Paperboard for Testing.
T 403 M-53 Bursting Strength of Paper.
T 410 os-61 Basis Weight of Paper and Paperboard.
T 411 M-44 Thickness and Density of Paper.
T 414 M-49 Internal Tearing Resistance of Paper.
T 452 M-48 Brightness of Paper.

(TAPPI Standards are available from the Technical Association of the Pulp and Paper Industry, 360 Lexington Avenue, New York 17, New York).

3. REQUIREMENTS

3.1 Material. All materials shall be of high commerical grade, and shall be such as to produce charts suitable in every respect for the intended purpose.

3.1.1 Cardboard. Cardboard shall be white tag board, having the following physical requirements. The cardboard shall pass the tests specified in 4.4.1.

Property	Value
Basis Weight	24 x 36 - 600 M (plus or minus 5%)
Caliper	.0203 (plus or minus .001) inch
Mullen	250 p.s.i. minimum
Tear	650/880 minimum
Brightness	85 minimum
Stock	100% bleached Kraft

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3.1.2 Ink. Ink shall be a jet black offset ink, suitable for printing on the cardboard specified in 3.1.1. The ink shall be capable of withstanding dry mounting process so that resulting letters are jet black and free of pinholes.

3.1.3 Eyelets. Eyelets shall be 1/4 inch diameter, fabricated of nickel plated-brass.

3.1.4 Spiral wire binding. The wire used for binding the two parts of each chart form shall be 0.0285-inch-diameter, tinned steel wire.

3.2 General description. The vision test chart set shall consist of three alternate forms of clinical and screening distant vision acuity test charts. Each chart form shall consist of two sections (charts), bound together with the spiral wire specified in 3.1.4. Each finished chart form shall measure approximately 22-1/2 inches wide by 23-1/2 inches long overall when the chart form is open and shall fold at the spiral binding vertically into two sections of equal size, so that when closed, each finished chart form, excluding binding, shall measure approximately 11-1/4 inches wide by 23-1/2 inches long. Each section of each chart form shall be made from three layers of cardboard, as specified in 3.1.1, laminated by means of a dry mounting process. Each complete chart form shall have sixteen lines of test letters. Printing shall be by means of an offset process using the master negatives loaned by the Government (see 3.4), to produce the printing plates. Each chart form shall be provided with two eyelets, as specified in 3.1.3, one in the top center of each section of the chart form, in the positions indicated by center lines on the master negatives. The two sections of each chart form shall be hinged by means of a spiral binding as specified in 3.1.4, approximately 7/16 inch outside diameter, and approximately 6 coils per inch. The outside corner of each chart form section shall be rounded on approximately 1/4 inch radius.

3.3 Test letters. The test letters shall be jet black. The contrast of letters on the cardboard shall be equal in intensity to test letters on the standard sample test charts, which will be loaned to the contractor by the Government. The test letters shall show sharp corners and edges. No fuzziness shall be detectable by an observer having 20/20 visual acuity when inspecting the charts from a distance of 5 inches.

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3.4 Master negatives. The Government will loan the contractor a set of master negatives consisting of a total of six negatives for the entire vision test chart set (one for each section of each chart form) sufficient to accomplish all printing. The master negatives shall be mated in pairs as follows to produce the three alternate chart forms:

Form No.	Line No. 1 Letter	Line No. 2 Letter
1	NS	ZVDSO
2	CZ	HZDCV
3	KD	CSVZN

3.5 Scoring instructions. The scoring instructions and applicable identifying form number are an integral part of the first half negative (lower left hand side) of each alternate chart form and shall be reproduced with the test letters on the front of the chart section.

3.6 Workmanship. Workmanship shall be first class throughout. Vision acuity testing chart sets shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. Sampling for examination shall be conducted in accordance with Standard MIL-STD-105.

4.2.1 Components. Sufficient samples of the components shall be selected from each lot of material used in the manufacture of the charts to perform tests as required by the applicable subsidiary specifications or by tests specified herein.

4.2.2 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The charts shall be examined to determine compliance with all requirements contained in this specification.

4.4. Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 The cardboard shall be tested in accordance with the following standards of the Technical Association of the Pulp and Paper Industry (TAPPI):

TAPPI Standard No.
T 401 M-53
T 402 M-49
T 403 M-53
T 410 os-61
T 411 M-44
T 414 M-49
T 452 M-48

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) chart set, as specified, constitutes one unit of issue.

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5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in procurement document.

Table I - Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	18 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each chart form shall be wrapped in kraft paper. Three (3) wrapped forms (1 unit) shall be packaged in a carton of suitable size and design or shall be wrapped in flexible corrugated fiberboard. Wrappings and carton shall be secured.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with Specification PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

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5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix of Specification PPP-B-636. In addition, waterproofing shall be as specified in 30.4.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE : Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to Specification PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with Specification PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class Domestic. Closure of fiberboard boxes shall be in accordance with Method I of Appendix to box specification.

5.3 Marking.

5.3.1 Unit packages. Each unit package shall be marked in accordance with Standard MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with Standard MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 The cardboad material referenced in 3.1.1 is Riegel Paper Corp.'s Tufwite.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army. - MD
Navy. - MS
Air Force. - 03

Review activities:
Army. - MD
Navy. - MS
Air Force. - 03

Category: C

Preparing activity:
Defense Supply Agency. - DM

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

FOLD UNDER

DEFENSE SUPPLY AGENCY
DEFENSE MEDICAL SUPPLY CENTER
3RD AVENUE AND 29TH STREET
BROOKLYN, N.Y., 11232

OFFICIAL BUSINESS

POSTAGE FEES PAID
DEFENSE SUPPLY AGENCY

DEFENSE SUPPLY AGENCY

DEFENSE MEDICAL SUPPLY CENTER

ATTN: DIRECTORATE OF TECHNICAL OPERATIONS (CODE T-15)

3RD AVENUE AND 29TH STREET

BROOKLYN, N.Y., 11232

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION MIL-C-36228A(DM), DATED 28 DECEMBER 1964. CHART SET, VISION ACUITY TESTING, CLINICAL AND SCREENING, DISTANT VISION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1426

SND PP30 12502

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